

Ashutosh Pattnaik

W340 Westgate Building
University Park, PA 16802

Email: ashutosh13@gmail.com
Homepage: <https://ashutoshpattnaik.github.io>

RESEARCH INTERESTS

GPU Architectures, Computer Graphics, CPU-GPU Architectures, Near-Data Computing

EDUCATION

Doctor of Philosophy in Computer Science and Engineering 08/25/2013 - 12/21/2019
Pennsylvania State University, University Park, PA
GPA: 3.82/4.0

Bachelor of Technology (Honors) in Electronics and Instrumentation Engineering 07/2009 - 04/2013
National Institute of Technology, Rourkela, India
GPA: 9.24/10.0

WORK EXPERIENCE

Staff GPU Architect, Arm Inc. 04/01/2023 - Present

Senior GPU Architect, Arm Inc. 09/30/2019 - 03/31/2023

- Working on the architecture design and optimization of the Arm Mali GPU to improve performance, power and area. Specifically working on modifying the architecture and design of the frontend of the GPU pipeline to enable better future scalability, feature addition and improved performance and power efficiency for next generation GPU applications

Research Assistant, The Pennsylvania State University Summer 2014 - Fall 2019
Advisor: Chitaranjan Das and Mahmut Kandemir W209 Westgate Building, University Park, PA - 16802

- Understanding research issues and opportunities involved in bringing near-data computing paradigm to GPUs and CPU-GPU systems and optimizing the compute placement for improved performance and energy efficiency.
- Improving the GPU datapath for improved execution of irregular applications and machine learning applications.

Co-Op Engineer, AMD Research 05/02/2016 - 08/26/2016
Mentors: Nuwan Jayasena, Yasuko Eckert *Manager: John Keaty* 1 AMD Place, Sunnyvale, CA - 94088

- Researched on efficient interconnect topologies and data placement techniques for a 3D-stacked processing-in-memory enabled, multi-chip module based architecture.

Co-Op Engineer, AMD Research 06/08/2015 - 09/04/2015
Mentor: Joseph Greathouse *Manager: John Keaty* 7171 Southwest Parkway, Austin, TX - 78735

- Developed methodology and micro-benchmarks to perform detailed characterization of the energy usage of different ISA instructions and data movement in AMD GPUs

Teaching Assistant, The Pennsylvania State University Fall 2013, Spring 2014
Department Head: Raj Acharya W209 Westgate Building, University Park, PA - 16802

- Teaching assistant for CMPEN 270: Digital Design: Theory and Practice in Fall 2013.
- Teaching assistant for CMPEN 431: Introduction to Computer Architecture in Spring 2014.

PUBLICATIONS

[MIDDLEWARE 2024] Vivek M. Bhasi, Aakash Sharma, Rishabh Jain, Jashwant Raj Gunasekaran, **Ashutosh Pattnaik**, Mahmut Taylan Kandemir, Chita R. Das, "Towards SLO-Compliant and Cost-Effective Serverless Computing on Emerging GPU Architecture", In the Proceedings of the 25th International Middleware Conference, Hong Kong, December 2024.

[ICCAD 2021] Weizheng Xu, **Ashutosh Pattnaik**, Geng Yuan, Yanzhi Wang, Youtao Zhang, Xulong Tang, "ScaleDNN: Data Movement Aware DNN Training on Multi-GPU", In the Proceedings of the 2021 IEEE/ACM International Conference On Computer Aided Design, November 2021. *Acceptance Rate $\approx 24\%$*

[ISCA 2020] Sonali Singh, Anup Sarma, Nicholas Jao, Ashutosh Pattnaik, Sen Lu, Kezhou Yang, Abhronil Sengupta, Vijaykrishnan Narayanan, Chita Das, "**NEBULA: A Neuromorphic Spin-Based Ultra-Low Power Architecture for SNNs and ANNs**", In the Proceedings of the 47th International Symposium on Computer Architecture, Worldwide Event, June 2020. *Acceptance Rate $\approx 18\%$*

[MICRO 2019] Haibo Zhang, Shulin Zhao, Ashutosh Pattnaik, Mahmut T. Kandemir, Anand Sivasubramaniam and Chita R. Das, "**Distilling the Essence of Raw Video to Reduce Memory Usage and Energy at Edge Devices**", In the Proceedings of the 52nd International Symposium on Microarchitecture, Columbus, Ohio, October 2019. *Acceptance Rate $\approx 23\%$*

[ISCA 2019] Ashutosh Pattnaik, Xulong Tang, Onur Kayiran, Adwait Jog, Asit Mishra, Mahmut T. Kandemir, Anand Sivasubramaniam and Chita R. Das, "**Opportunistic Computing in GPU Architecture**", In the Proceedings of the 46th International Symposium on Computer Architecture, Phoenix, Arizona, June 2019. *Acceptance Rate $\approx 17\%$*

[MEMSYS 2019] Anup Sarma, Huaipan Jiang, Ashutosh Pattnaik, Jagadish Kotra, Mahmut T. Kandemir, Chita R. Das, "**CASH: Improving DRAM Energy Efficiency in CPU-based Inference**", In the Proceedings of the 5th International Symposium on Memory Systems, Washington D.C., October 2019.

[SIGMETRICS 2019] Xulong Tang, Ashutosh Pattnaik, Onur Kayiran, Adwait Jog, Mahmut T. Kandemir and Chita R. Das, "**Quantifying Data Locality in Dynamic Parallelism in GPUs**", In the Proceedings of the ACM on Measurement and Analysis of Computing Systems, Phoenix, Arizona, June 2019. *Acceptance Rate $\approx 8\%$*

[HPCA 2017] Xulong Tang, Ashutosh Pattnaik, Huaipan Jiang, Onur Kayiran, Adwait Jog, Sreepathi Pai, Mohamed Ibrahim, Mahmut T. Kandemir and Chita R. Das, "**Controlled Kernel Launch for Dynamic Parallelism in GPUs**" In the Proceedings of the 23rd International Symposium on High Performance Computer Architecture, Austin, Texas, February 2017. *Acceptance Rate $\approx 22\%$*

[PACT 2016] Ashutosh Pattnaik, Xulong Tang, Adwait Jog, Onur Kayiran, Asit Mishra, Mahmut Kandemir, Onur Mutlu and Chita Das, "**Scheduling Techniques for GPU Architectures with Processing-In-Memory Capabilities**" In the Proceedings of the 25th Parallel Architecture and Compilation Techniques, Haifa, Israel, September 2016. *Acceptance Rate $\approx 22.3\%$*

[PACT 2016] Onur Kayiran, Adwait Jog, Ashutosh Pattnaik, Rachata Ausavarungnirun, Xulong Tang, Mahmut Kandemir, Gabriel Loh, Onur Mutlu and Chita Das, " **μ C-States: Fine-grained GPU Datapath Power Management**" In the Proceedings of the 25th Parallel Architecture and Compilation Techniques, Haifa, Israel, September 2016. *Acceptance Rate $\approx 22.3\%$*

[IISWC 2016] Vignesh Adhinarayanan, Indrani Paul, Joseph Greathouse, Wei N. Huang, Ashutosh Pattnaik and Wuchun Feng, "**Measuring and Modeling On-Chip Interconnect Power on Real Hardware**", In the Proceedings of IEEE International Symposium on Workload Characterization, Providence, Rhode Island, 2016. (**Best Paper Award**) . *Acceptance Rate $\approx 30.4\%$*

[SIGMETRICS 2016] Adwait Jog, Onur Kayiran, Ashutosh Pattnaik, Mahmut Kandemir, Onur Mutlu, Ravi Iyer and Chita Das, "**Exploiting Core-Criticality for Enhanced Performance in GPUs**", In the Proceedings of the 42nd ACM International Conference on Measurement and Modeling of Computer Systems, Antibes Juan-les-Pins, France, June 2016. *Acceptance Rate $\approx 13.4\%$*

[MEMSYS 2015] Adwait Jog, Onur Kayiran, Tuba Kesten, Ashutosh Pattnaik, Evgeny Bolotin, Nilardish Chatterjee, Steve Keckler, Mahmut Kandemir and Chita Das, "**Anatomy of GPU Memory System for Multi-Application Execution**", In the Proceedings of the 1st International Symposium on Memory Systems, Washington D.C., October 2015.

[ICCCS 2012] Ashutosh Pattnaik, Sharad Agarwal, Subhasis Chand, "**A New and Efficient Method for Removal of High Density Salt and Pepper Noise Through Cascade Decision based Filtering Algorithm**", In the Proceedings of the 2nd International Conference on Communication, Computing & Security, India, 2012.

HONORS AND AWARDS

Best Paper Award:

- Measuring and Modeling On-Chip Interconnect Power on Real Hardware, IISWC 2016

Student Travel Grant:

- 2019: IEEE Travel Grant for ISCA, NSF Travel Grant for SIGMETRICS
- 2017: IEEE Travel Grant for HPCA, ACM SIGARCH Travel Grant for ISCA
- 2016: ACM SIGMETRICS Travel Grant, NSF Travel Grant for PACT
- 2015: ACM SIGARCH Travel Grant for ISCA

SERVICE

Conference Organization

- **Workshops & Tutorials Co-Chair**, ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Pittsburgh, PA, March 2026
- **General Co-Chair**, Workshop on General Purpose Processing Using GPU (GPGPU), San Diego, CA, February, 2020
- **Submission/Web Chair**, Workshop on General Purpose Processing Using GPU (GPGPU), Providence, RI, April 2019
- **Submission Chair**, International Conference on Supercomputing (ICS), Turkey, June 2016

Conference Session Chair

- **High Stakes Vision - 1**, IEEE International Symposium on High-Performance Computer Architecture, Las Vegas, NV, March, 2025
- **GPUs**, IEEE International Symposium on High-Performance Computer Architecture, San Diego, CA, Feb, 2020

Program Committee Member

- IEEE International Symposium on High-Performance Computer Architecture, Sydney, Australia, February, 2026
- IEEE International Symposium on High-Performance Computer Architecture, Las Vegas, NV, March, 2025

External Review Committee Member

- IEEE/ACM International Symposium on Microarchitecture, Seoul, Korea, October, 2025
- IEEE/ACM International Symposium on Microarchitecture, Chicago, IL, October, 2022
- ACM International Symposium on Computer Architecture, New York City, NY, June, 2022
- IEEE/ACM International Symposium on Microarchitecture, Virtual Event, October, 2021
- IEEE/ACM International Symposium on Microarchitecture, Virtual Event, October, 2020
- IEEE International Symposium on High-Performance Computer Architecture, San Diego, CA, Feb, 2020

Journal Reviewer:

- Transactions on Cloud Computing, IEEE
- Transactions on Parallel and Distributed Systems, IEEE
- Transactions on Architecture and Code Optimization (TACO), IEEE
- Transactions on Computers (TC), IEEE
- Microprocessors and Microsystems: Embedded Hardware Design, Elsevier
- ETRI Journal, Wiley

REFERENCES

References are available on request.